

PROBLEM STATEMENT 1 OF 2

Campus Buzz — The Campus Coordination Platform

Campus life coordination is fragmented — scattered WhatsApp groups, physical notice boards, and word-of-mouth. There is no verified, students-first social layer built for college life.

Build one. A mini social platform for your campus where students coordinate, clubs communicate, events are discovered, and concerns are raised. Verified identities, real-time interaction, and a feed that actually reflects campus life.

1.1 Authentication & Roles

Login via roll number and institute email. On successful verification, a role is assigned to the account. All interactions on the platform are role-gated at the server level.

ROLE

Student

Any verified student. Can post to Campus Buzz, join coordination rooms, raise complaints, and view events.

ROLE

Club / Committee

Verified org accounts. Can post official content, create events on the calendar, and embed forms in posts.

ROLE

Admin

Platform-level control. Moderates content, manages users, resolves escalated complaints, creates events on behalf of students.

Campus Buzz Feed

The default landing feed. Any verified student can post. Posts are the primary coordination primitive on the platform.

POST STRUCTURE

Every post must include: an image, a title, a description, and at least one hashtag — the hashtag is mandatory and drives intent classification. The feed is filterable by hashtag.

HASHTAG INTERACTION MODEL

The action triggered when a viewer taps a post is determined entirely by its hashtag. Different hashtags produce different interaction surfaces.

HASHTAG	ON TAP — WHAT HAPPENS
<code>#foodsplit</code>	Opens a live group chat room unique to that post. Students coordinate their order in the room. Poster closes the room once done.
<code>#cabsplit</code>	Opens a live group chat room unique to that post. Students headed the same way join to coordinate and split the fare.
<code>#resell</code>	Opens a live group chat room unique to that post. Interested buyers join; negotiation and payment happen in person. Poster closes when sold.
<code>#lost</code> / <code>#found</code>	Shows the poster's contact information directly — no group chat room. Coordination happens privately.

POST EXPIRY

For `#foodsplit` and `#cabsplit` posts, a timer is mandatory:

- Default lifespan is **24 hours** — the post auto-deletes when the timer ends.
- The poster may set a custom timer ranging from **10 minutes to 2 days**.
- The remaining timer must be visible on the post card in the feed.
- Auto-deletion must be handled by a **background job** — not inline with API requests.

Club / Committee & Admin Section

A separate feed section accessible only to verified Club, Committee, and Admin role accounts. This is not part of the student Campus Buzz feed — it is a distinct, role-restricted space for official communication.

- Posts are free-form — text, links, and embedded Google Forms are supported. No hashtag requirement.
- These posts may optionally be linked to an event on the Event Calendar, surfacing both in the section and on the calendar view.
- Posting to this section must be **blocked at the API level** for student role accounts. The club/committee distinction from the student feed must be enforced server-side.

Event Calendar

A shared, campus-wide event calendar. All students see upcoming events from clubs, committees, and admin. The calendar is a read view for students and a management surface for clubs, committees, and admins.

- Clubs and committees can **create, edit, and manage** their own events directly.
- Students can **request an Admin** to create an event on their behalf — a lightweight request workflow.
- Events display: name, date, time, venue, and description.
- Events created from the Club/Committee section posts are linked and shown on the calendar.

Complaints

Any verified student can raise a complaint visible to the campus community. The purpose is community awareness and visibility — not a formal ticketing workflow with an external resolution authority.

- Complaints are **anonymous** — the poster's identity is not exposed in the public view. The backend must store the real user identity internally.
- The original poster can **mark their own complaint as resolved** once the issue has been addressed.
- Admin can view and moderate all complaints, including the real poster identity at the admin level.

Open Scope

The features above define the floor — the minimum expected surface area. Teams are encouraged to go beyond. Your score benefits from depth and creativity within the scope, not just checklist completion.

Ideas to Consider

- Tag suggestions and trending hashtags in the feed
- Smart post expiry — notify poster before auto-deletion
- RSVP and event reminders on the calendar
- Push or in-app notification system for active chat rooms
- Moderation, reporting, and trust systems for posts and users
- Anything else that makes campus life genuinely better

This is your campus. Build accordingly.

Evaluation — Problem Statement 1

FEATURE COMPLETENESS

All core features implemented and working — feed, hashtag interactions, group chat rooms, calendar, complaints.

TECHNICAL DEPTH

Real-time state, background jobs for post expiry, server-enforced access control, anonymous posting handled correctly.

ARCHITECTURE & CODE QUALITY

Schema design, normalisation, role gating at the server level. Can every member explain every part of the codebase?

UX & OPEN SCOPE

Clarity of use across all three roles. Creativity and polish in open scope additions.

PROBLEM STATEMENT 2

Problem Statement 2 — AI Recruitment Pipeline

PROBLEM STATEMENT 2 OF 2

AI Recruitment & Screening Platform

Campus recruitment is manual, inconsistent, and slow. Shortlisting is gut-feel. Interviewers walk in blind. Candidates wait days for results that arrive through a WhatsApp forward. The process repeats every semester, and it never improves.

Build a platform that fixes this — end to end. A structured, AI-assisted recruitment pipeline where every stage is tracked, every score is justified, and every candidate gets a fair shot.

ROLE

Admin

Creates the org, opens recruitment drives, sets stage weightages, assigns interviewers, and makes final selections.

ROLE

Applicant

Registers with email and password. Discovers orgs, applies to drives, completes the AI round, and tracks their status.

ROLE

Interviewer

Assigned by admin. Views candidate profiles, AI scores, and Q&A. Submits a structured scorecard after conducting the offline interview.

All three roles must be enforced via server-side middleware. No role may access another role's endpoints regardless of what the client sends.

Recruitment Pipeline

Each recruitment drive follows a strict linear pipeline. Every stage gates the next — the platform must enforce stage transitions at the API level.



1 Org Setup — Admin

Admin creates the org profile: name, description, logo, social links, and domains or departments (e.g. Tech, Design, Marketing).

Admin opens a **Recruitment Drive**: role name, description, eligibility criteria, application deadline. Stage weightage configuration is set per drive — for example, CV Score: 20%, AI Round: 40%, PI Round: 40%. Weightages must sum to 100%. Multiple drives can run simultaneously under one org, each with independent applicant pools, interviewers, and timelines.

2 Discovery & Apply — Applicant

Applicant lands on the platform and sees all orgs and their active drives. They open a role, read the description and eligibility, and click Apply.

The application form collects: name, email, year, branch, and a CV uploaded as PDF. Applications are locked after the deadline — no edits or late submissions are permitted. The PDF must be stored in cloud storage (Supabase Storage or Cloudinary); the database stores only the file URL.

3 AI Screening — Automated, triggered on deadline close

The AI screening flow runs in two complementary passes for each candidate:

CV Scoring: The system parses the uploaded CV — extracting skills, projects, experience, and achievements. AI evaluates the extracted profile against the role description and produces a **CV match score** measuring skill relevance, project alignment, and experience fit. This score contributes to the candidate's final score at the configured CV weightage.

AI Interview Round: Using the parsed CV and role description, the AI generates 5–8 personalised, role-specific questions unique to each candidate — no two candidates receive the same set. The applicant receives a timed window to attempt the round. Questions appear one

by one; the applicant types responses. On submission, AI scores each response and produces an **AI round score**.

Both scores are stored against the candidate's application and are visible to admin and assigned interviewers. AI integration is mandatory — use the Claude API, OpenAI API, Gemini API, or any open-source model. Hardcoded question banks are not accepted.

The AI screening flow must be offloaded to a **background job queue** — it must not run inline with the HTTP request that triggers it.

4 PI Round — Interviewer

Admin advances the drive to the PI Round stage and assigns interviewers to candidates or domains. An interviewer can only access candidates in drives they have been explicitly assigned to — this must be enforced server-side.

Interviewer opens a candidate and sees: full profile, uploaded CV link, AI-generated questions, the candidate's responses, and the AI score. After conducting the offline personal interview, the interviewer returns to the platform and submits a **structured scorecard** — per-criteria scores and written remarks. PI round scores are locked once submitted.

5 Score Matrix & Selection — Admin

All three stage scores (CV Score, AI Round Score, PI Score) are aggregated using the admin-configured weightages into a **final score** per candidate. The aggregation must happen at the database or application layer — not on the frontend.

Admin and interviewers see a full matrix: rows are candidates, columns are CV Score / AI Round Score / PI Score / Final Score. The matrix is sortable and filterable by any column or domain.

Adjusting stage weightages after scores are submitted must automatically recompute all rankings without data loss. Admin marks each candidate as **Selected**, **Waitlisted**, or **Rejected**.

6 Automated Result Notifications

On admin marking selection statuses, the platform automatically dispatches result emails to all candidates in a single operation. Emails are sent programmatically via Resend or Nodemailer — no manual sending.

Selected candidates receive an acceptance email. Waitlisted candidates receive a waitlist email. Rejected candidates receive a professional thank-you email. Email templates are pre-configured per drive. Failed send attempts must be logged and visible as a per-candidate delivery status report in the admin view.

2.3 Org Hub

Separate from the recruitment pipeline — the org's public-facing presence page. The Org Hub remains live outside of active recruitment seasons.

- Public page visible to anyone — unauthenticated visitors included.
 - Displays: org name, description, social links, active and past recruitment drives, announcements, and upcoming events.
 - Admin can post announcements and create events with date, time, and description.
 - Applicants and visitors can register for events directly from the Org Hub.
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Open Scope

The pipeline above defines the floor — the minimum expected system. Teams are encouraged to extend it. Depth and creativity within the scope are rewarded.

Ideas to Consider

- Applicant dashboard — track application status across all stages in real time
- Admin analytics — drop-off rates, score distributions, domain-level insights
- Interviewer scheduling — calendar integration for PI slot booking
- Multi-round AI screening — eliminate weak candidates before generating interview questions
- Anti-cheat mechanisms for the AI round window
- Collaborative scoring — multiple interviewers per candidate with averaged scores
- Anonymous percentile rank shown to applicants after results

Build the recruitment experience you wish you had.

Evaluation — Problem Statement 2

FEATURE COMPLETENESS

All pipeline stages implemented and gate each other correctly — apply, AI screen, PI round, scoring, email dispatch.

AI INTEGRATION QUALITY

CV parsing accuracy, question personalisation per candidate, scoring rubric quality. No hardcoded question banks.

SYSTEM DESIGN

Schema, role enforcement across all three roles, background queue for AI processing, weighted score aggregation at the data layer.

UX & OPEN SCOPE

Experience across all three roles — Admin, Applicant, Interviewer. Depth and creativity of open scope features.

Submission Requirements

All submission materials must be ready before **11:59 PM IST on 27 June 2026**. Late submissions are not accepted under any circumstance.

1 GitHub Repository

A single public GitHub repository containing the complete codebase on the `main` branch. Must include a `README.md` with setup instructions, environment variable requirements (a `.env.example` is strongly recommended), and a brief note on the team's architecture decisions.

2 Video Demo

A screen-recorded video demonstrating all implemented features. Real-time features must be shown working live — not mocked or simulated. Share as a Google Drive link with access set to Anyone with the link.

3 Submission Form

Submit both the repository link and the video link through the official submission form. The form link will be shared before the build window opens. Submissions through any other channel will not be considered.

Tools & Resources — No Restrictions

Use everything available to you — AI assistants, Copilot, open-source boilerplate, libraries, documentation, senior advice. There is no restriction on resources. What is evaluated is whether the team understands what they have built. During evaluation, every member must be able to explain any part of the codebase — schema design, architectural choices, trade-offs, and engineering decisions.